AE 452/478 Soil and Water Engineering II (2,3,3)

Lecturer: Dr. Wilson A. Agyare

Course Content (1/3)

- Agrohydrology
  - definition and importance
  - hydrological cycle
    - Precipitation
    - evapotranspiration
    - Infiltration
    - Streamflow
    - Groundwater

Course Content (2/3)

- Transecting
- Simple surveying instruments
  - Line level
  - A-Frame
  - N-Frame
- Scale
- Soil conservation
  - definition and importance
  - mechanics and agents of erosion
    - types of erosion
    - erosivity and erodibility
    - soil loss prediction
    - USLE
  - principles of erosion control
    - agronomic and engineering methods

Course Goals

1. To revise with students the hydrological cycle and its components
2. To develop analytical skills relevant in the areas of precipitation, evapotranspiration, infiltration and streamflow
3. To develop basic practical skills in transecting and simple surveying for soil conservation, infiltration measurement, and scale
4. To introduce student to the processes and mechanics of erosion, its assessment and control
5. To introduce student to agronomic and mechanical soil and water conservation measures

Reference Material

1. Fundamentals of soil physics by Daniel Hillel
2. Soil and water conservation engineering by G. O. Schwab
3. Hydrology in practice by E. M. Shaw
4. Soil erosion and conservation by R. P. C. Morgan
5. Development of conservation farming on hillslopes by Moldenhauer *et al.*

Course Assessment

1. Minimum of two (2) Quiz
2. One (1) mid-semester exams
3. Project or practical work and assignments
4. One (1) End-of-semester examination

AE 452 Practicals

- Transecting
- Simple surveying methods
- Infiltration measurement?

Transecting

Learning Objective

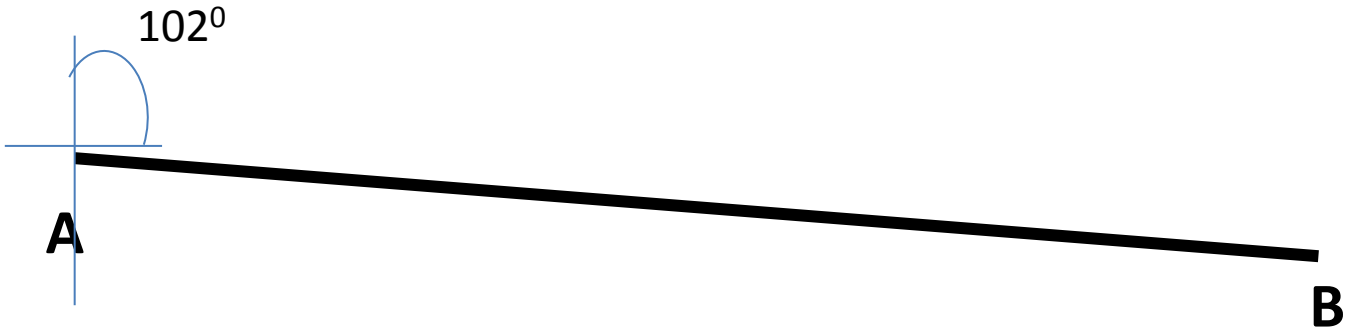
Students will be able to:

- Mark out baselines
- Mark out transects using the 3-4-5 method
- Avoid obstacles using the 3-4-5 method

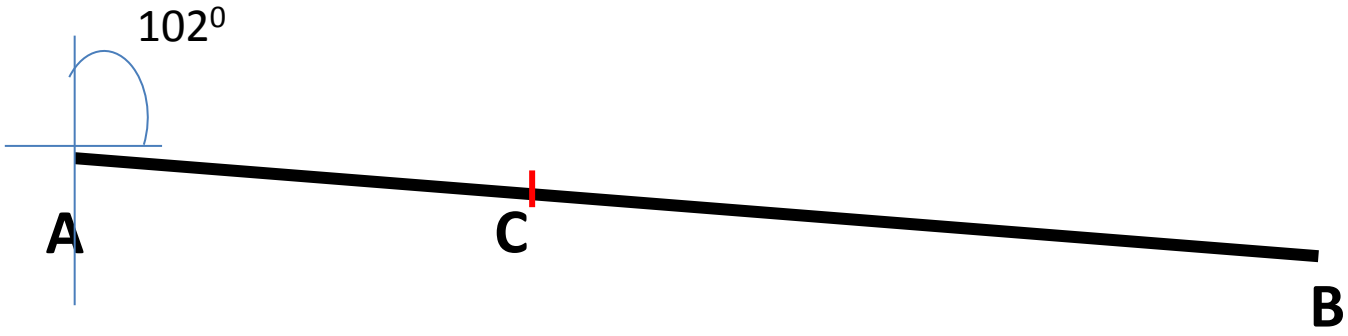
Transecting

- Setting out a baseline (compass/reference point, ranging poles, tape measure and chain)
- Setting out a transect perpendicular to the baseline (tape measure , poles, chain and the 3-4-5 method)
- Setting out a transect to avoid an obstacle (tape measure , poles, chain and the 3-4-5 method)

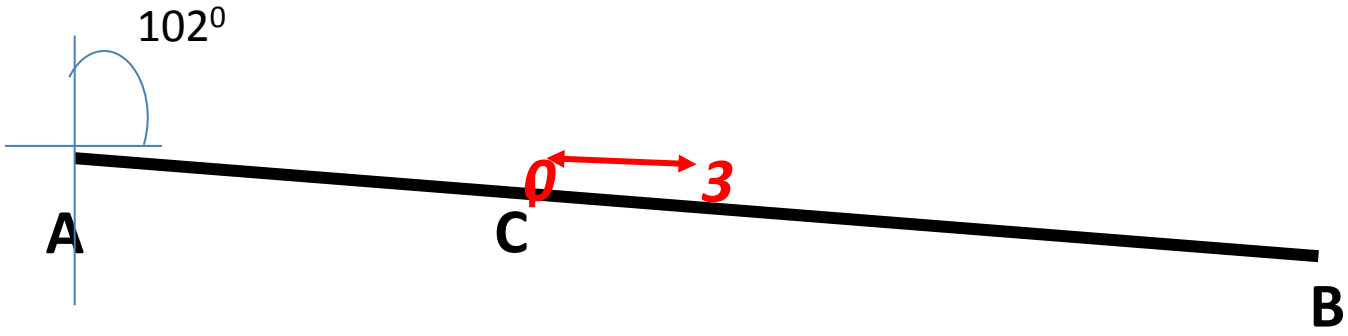
Baseline



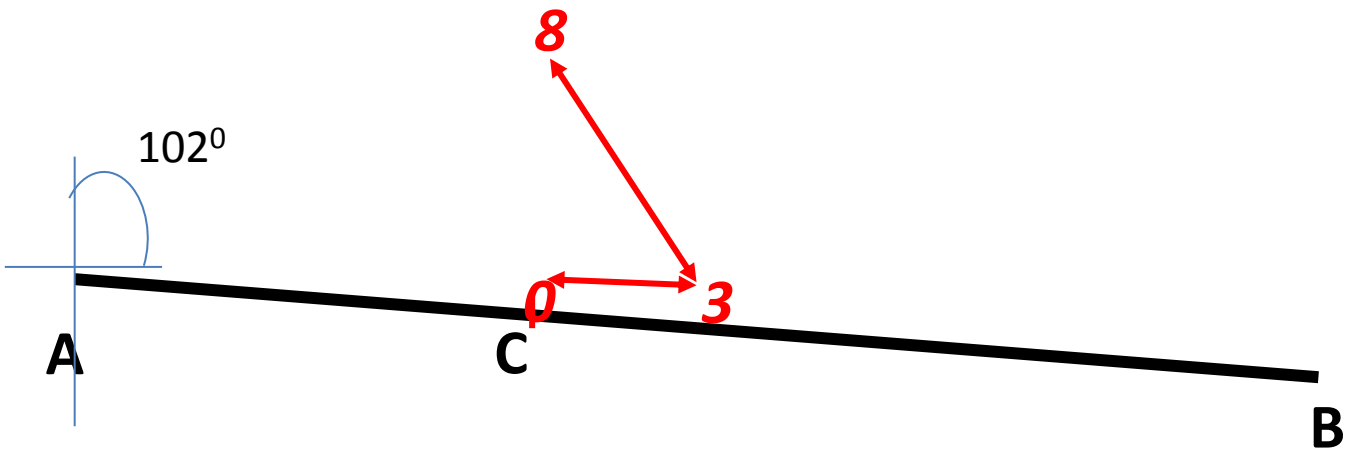
Transecting



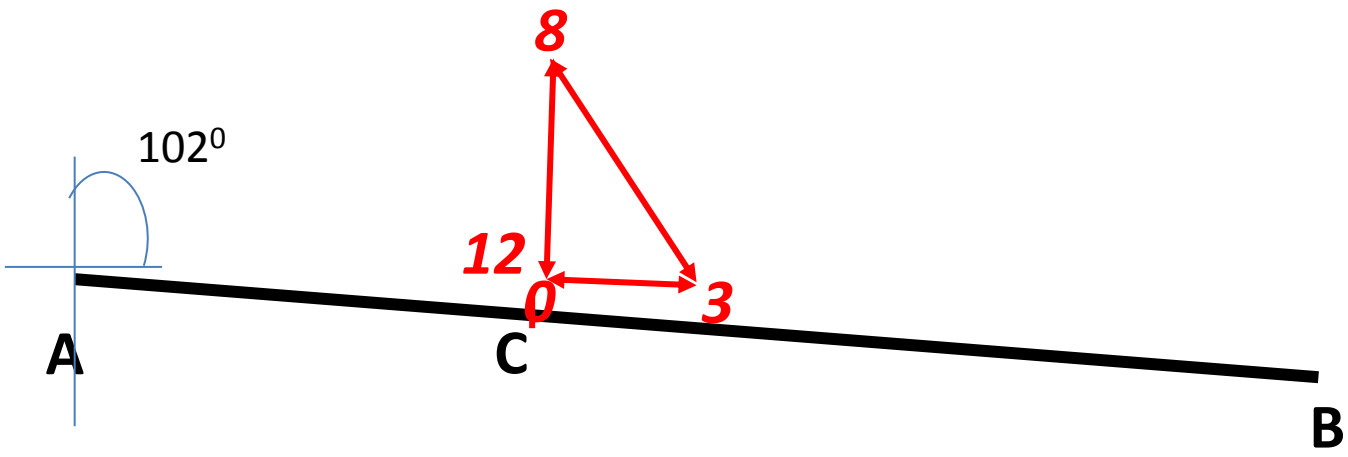
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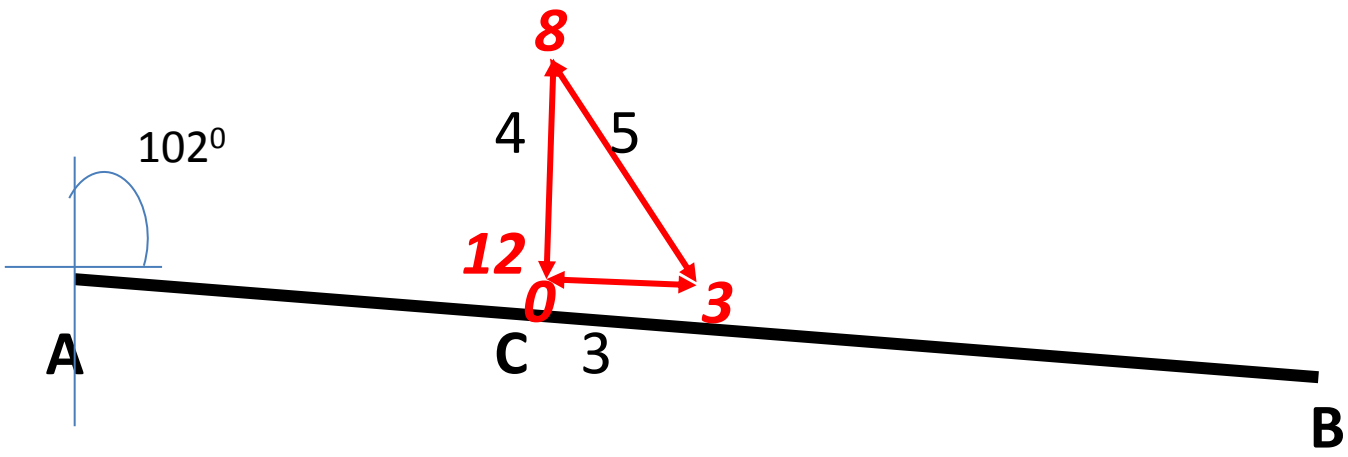
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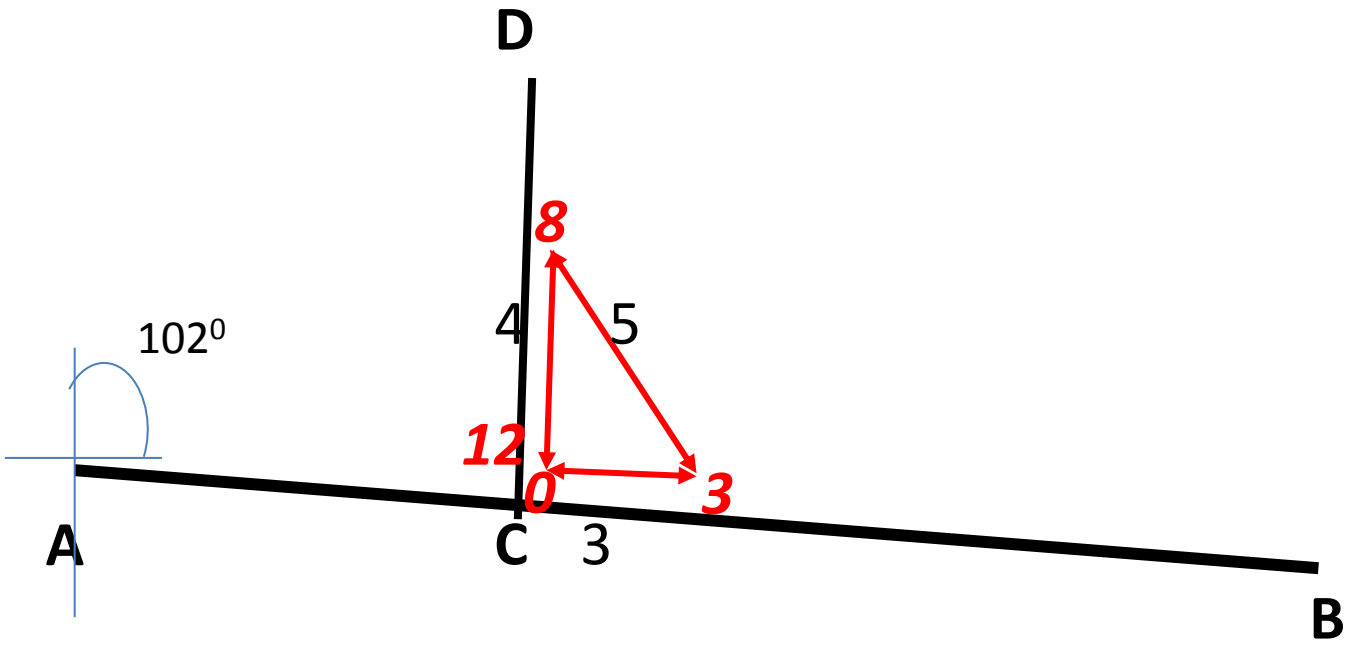
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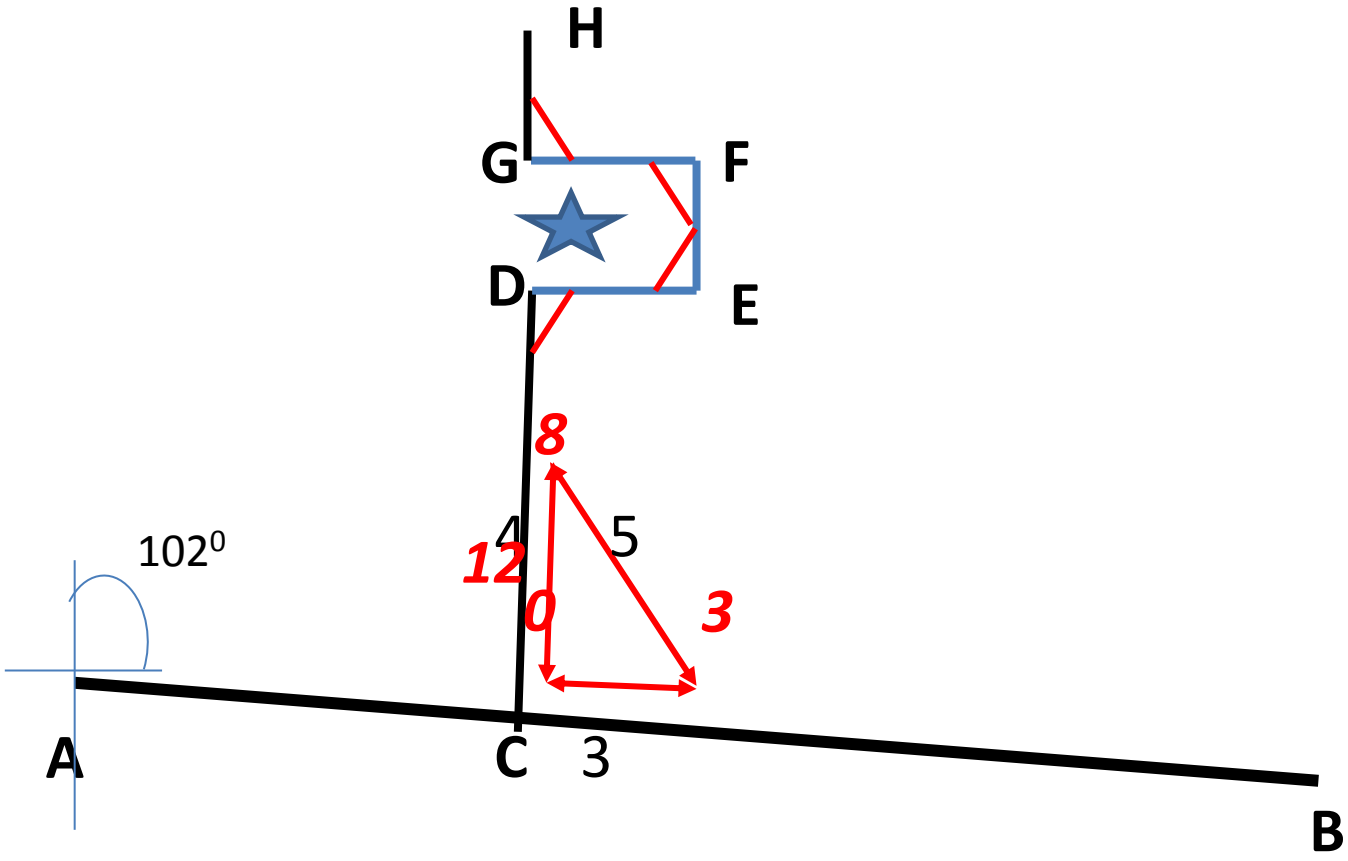
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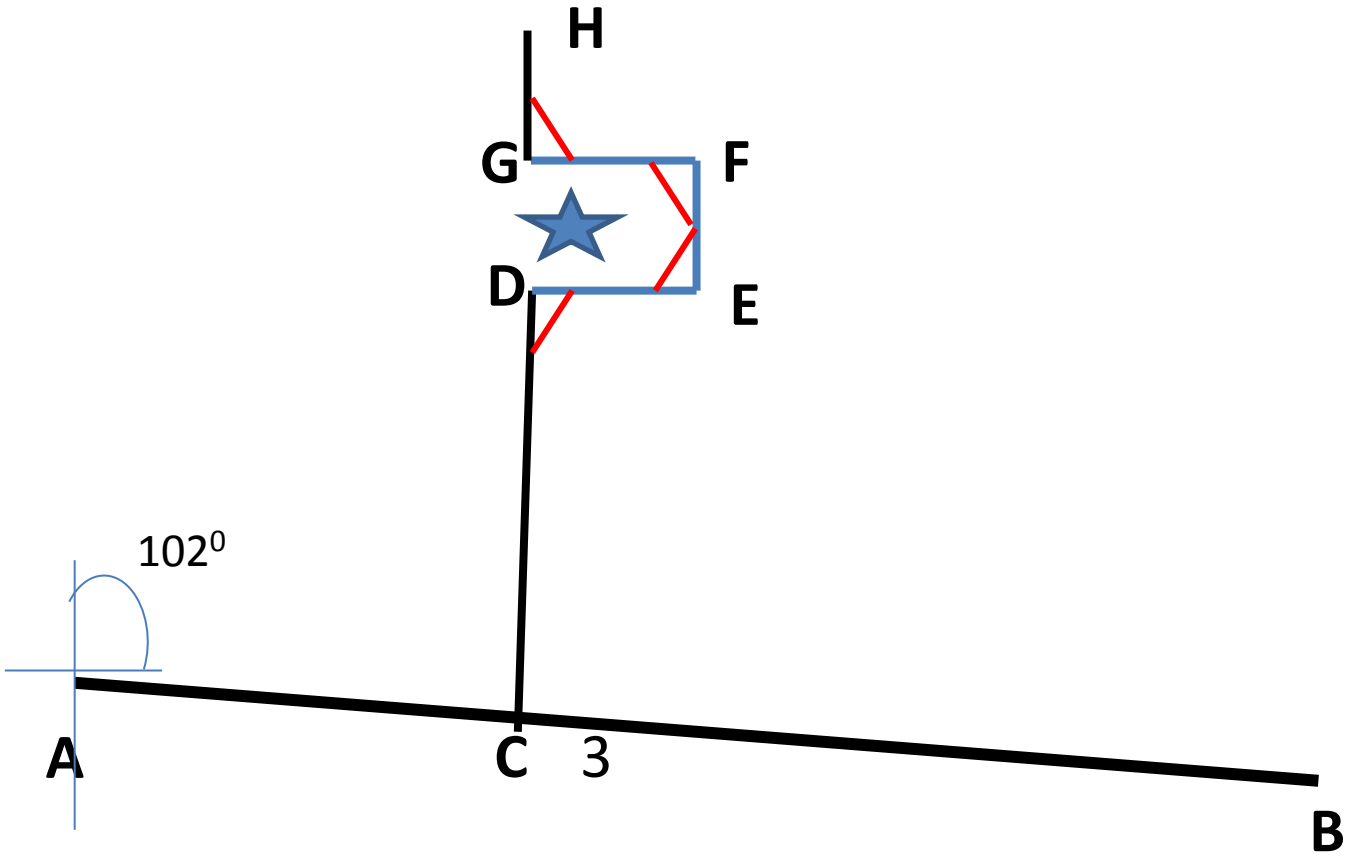
Transecting



Avoiding Obstacle



Avoiding Obstacle



Simple Surveying Instruments

Learning Objective

Students will be able to:

- construct and calibrate an A-frame and line level
- determine contours, channel slope of field and water courses using the A-frame and line level

Terminologies

For good soil water management it is important to understand the nature of the terrain one is working on by determining the contour to limit measures that will enhance soil and water loss

- **Contour:** Contours are imaginary lines joining points of equal height
- **Vertical Interval (VI):** Is the height difference or vertical distance between adjacent contours or terrace
- **Gradient**: Is the slope of the field or between contours

Terminologies

- **Surveying**: Field survey involves taking measurement from which maps and plans can be drawn.
  - The measurements taken include heights/elevations, distances and angles

- **Plan**: A plan is a drawing showing existing land and features with the proposed work or construction superimposed upon it.

- **Map**: Is a plan drawn to scale.

Simple Surveying Instruments

- Line level
- A-frame
- N-frame
- Liquid level

Line Level

What is a line level?

- A line level is a device constructed out of wooden poles, a twine and a spirit level to identify points of equal height or determine height difference on the earth.
- The device can be used to measure slope percentage or surface gradient.

Uses

- Contour
- Slope
- Measure height difference

Line Level

Materials required for line level

- two identical graduated sticks (1.5 m height graduated at 5 cm interval)
- non-elastic string (10 m)
- Spirit/string level

Line Level

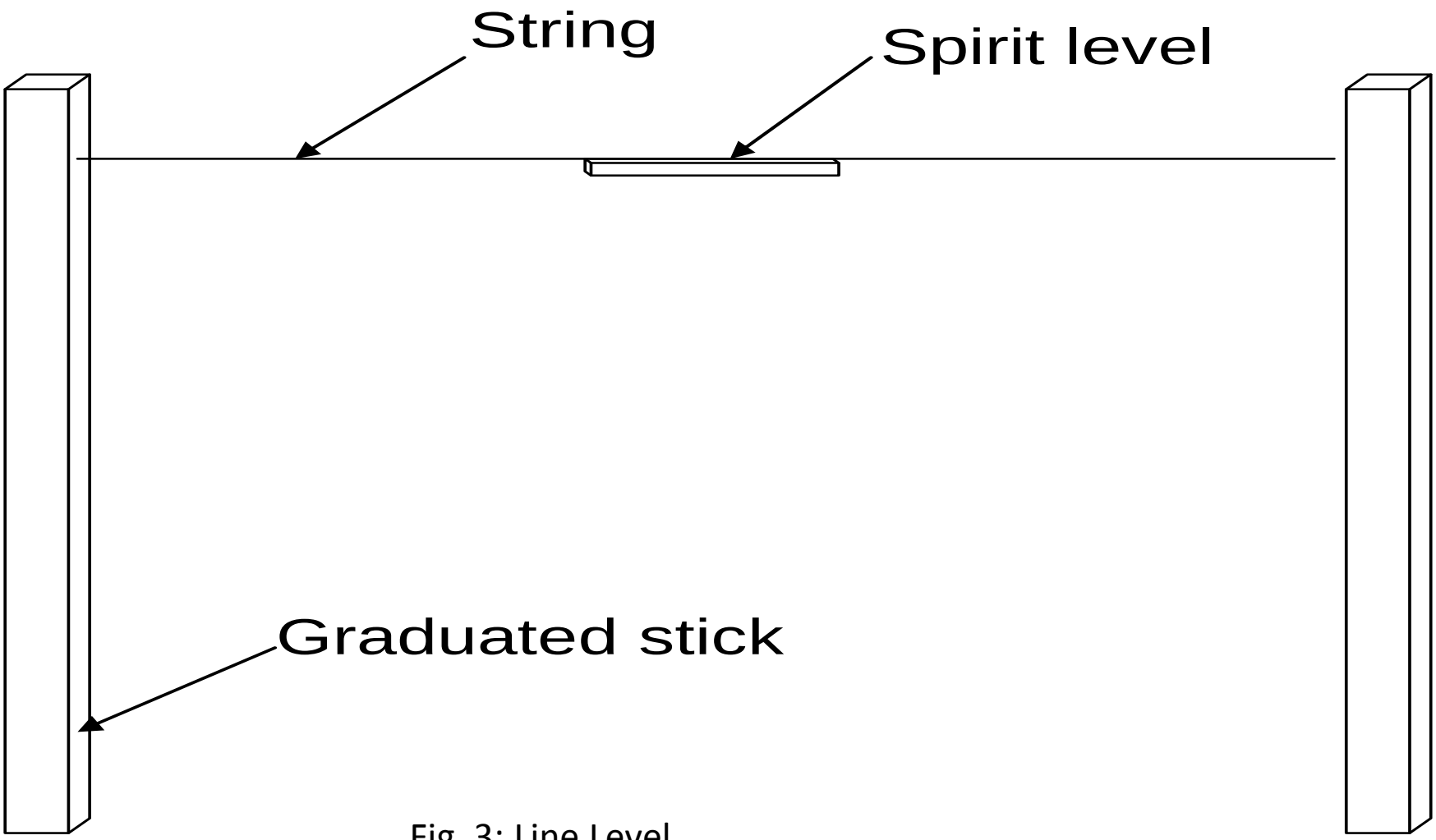
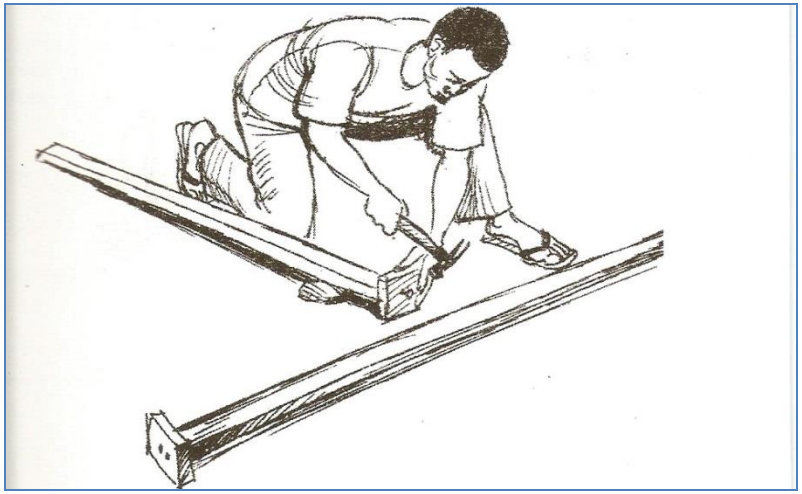
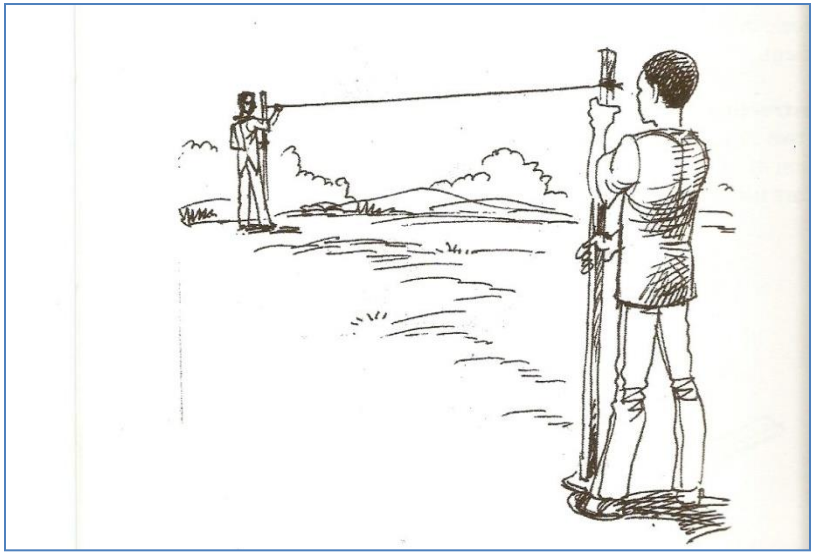


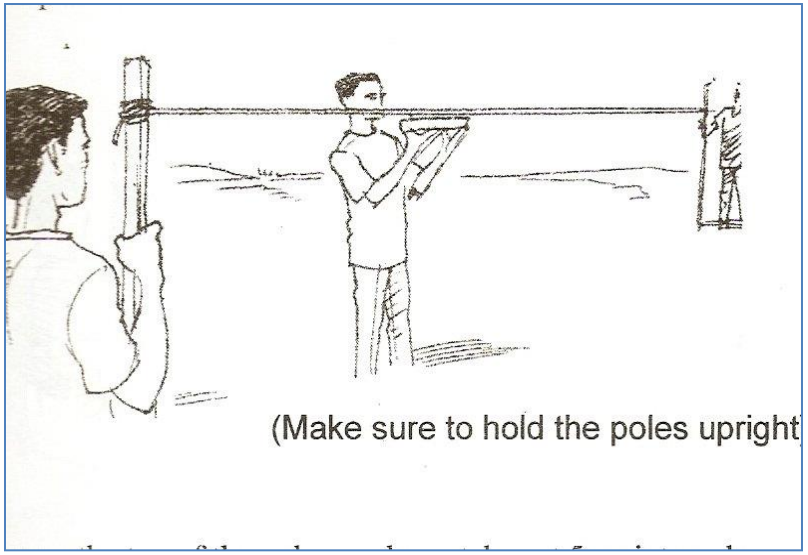
Fig. 3: Line Level

Constructing the line level









Line Level

Construction

1. The two sticks are graduated identically at 5 cm interval starting from the top to about 50 cm mark below.

2. The non-elastic string is then tied around the notches on each stick. The length of the string between the sticks is measured (e.g. 10 cm).

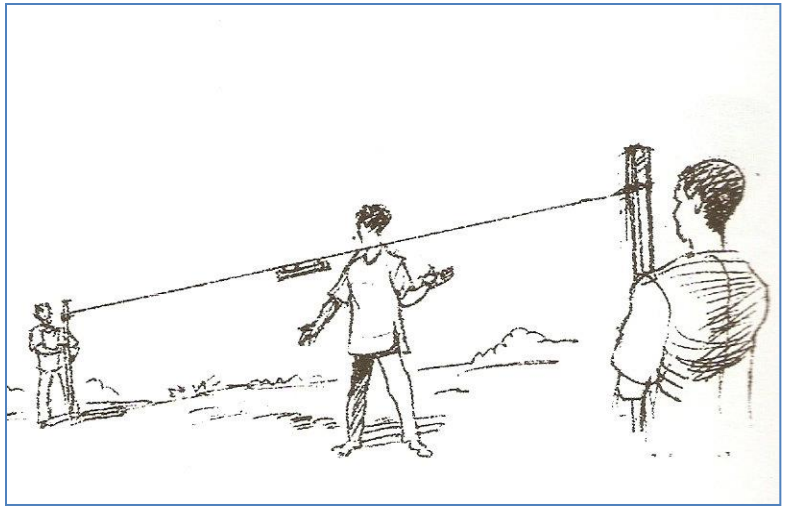
3. The centre of the string is marked and the spirit level suspended

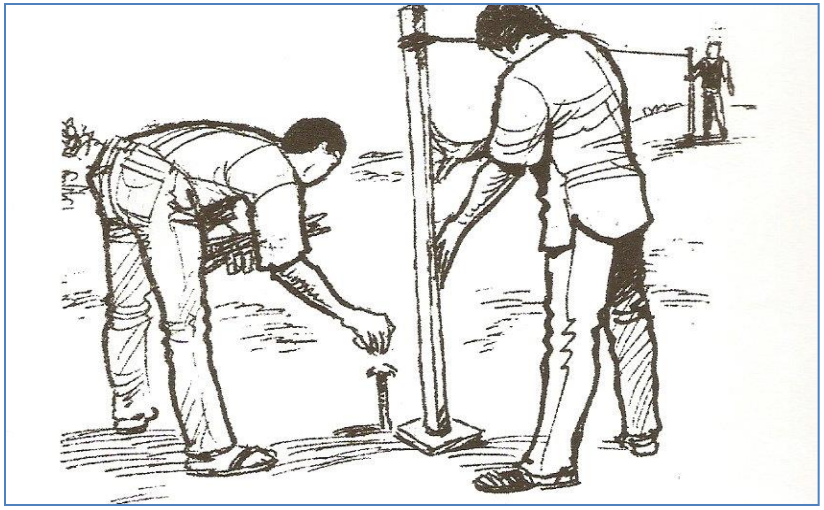
Line Level

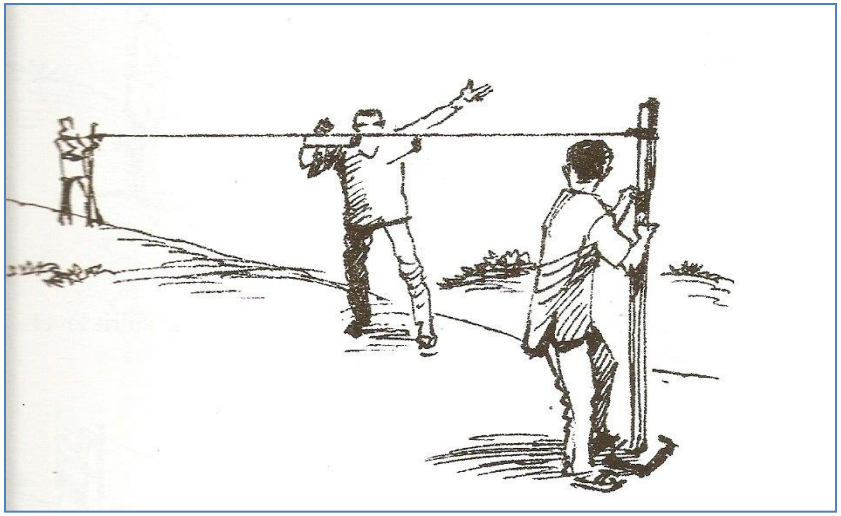
Setting out contour

- The two sticks are held vertically upright with the first stick placed at the starting point.
- With the string stretched tightly between the two sticks, the second stick is moved around by try and error to locate a position where the bubble in the spirit-level indicates level.
- This gives the location of equal height. The position is pegged and the procedure repeated.

Marking a contour with a line level







Line level

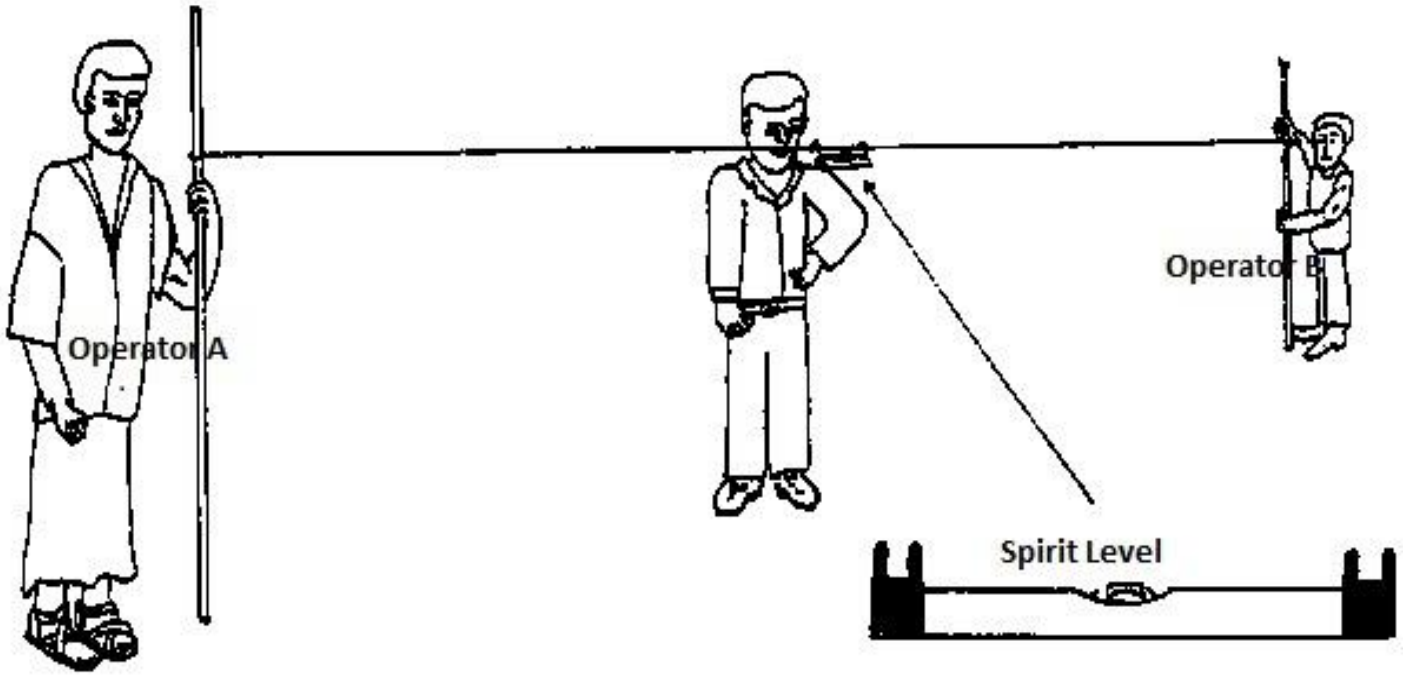


Fig. : Operation of a line level

Line Level

Marking out slope

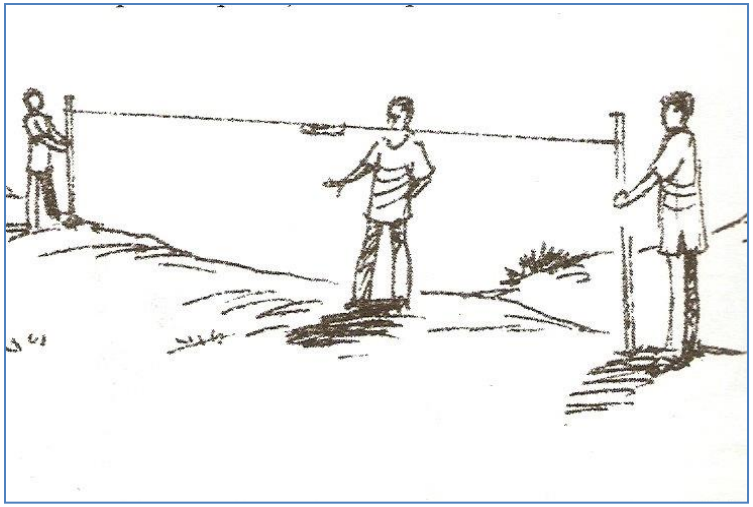
1. The sticks are held vertically apart with the string stretched.
2. The spirit level is positioned exactly in the middle of the string.
3. Operator **A** stands exactly upslope of Operator **B** and adjusts the string to get the bubble indicate perfectly levelled.

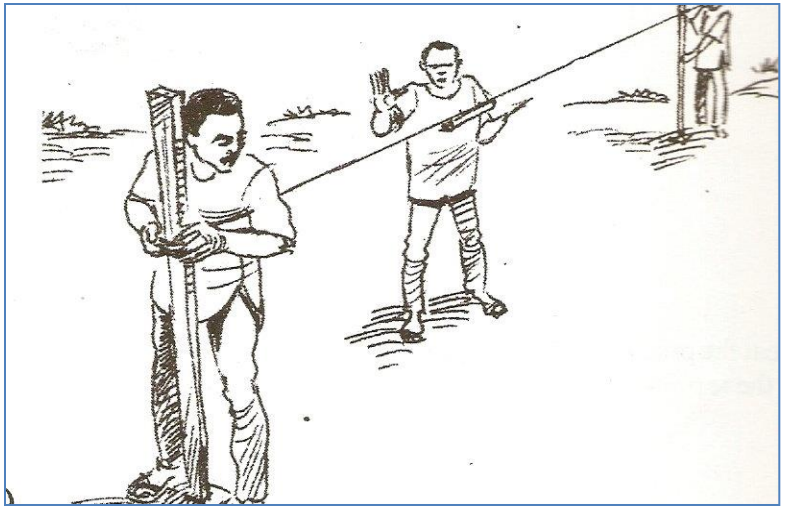
Line Level

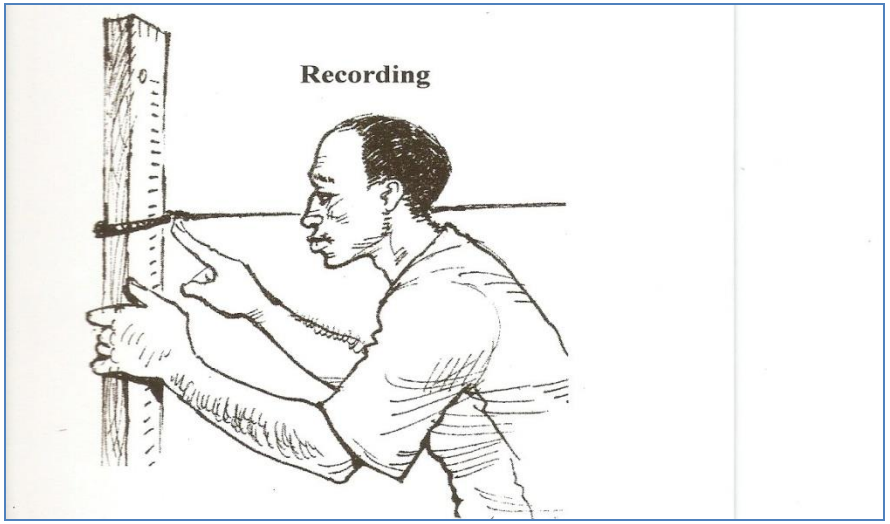
Marking out slope

4. When the bubble in the level is between the two marks this means that the sticks are positioned horizontally.
5. The height of the sticks is read which represents the difference in height between the two levelling sticks.

Measuring percentage slope of gradient







Line level

Calculation:

$$Slope = \frac{Vertical\ Height\ difference}{Horizontal\ Distance} \times 100\%$$

Line Level

Table. Relation between height difference and slope for line level

Height difference (cm)	Slope (%) measured with string of	
	5 m length	10 m length
5	1.0	0.5
10	2.0	1.0
15	3.0	1.5
20	4.0	2.0
25	5.0	2.5
30	6.0	3.0

A-frame (1/11)

What is an A-frame?

- Is a device constructed out of two wooden poles, a cross bar and a plummet in the form of the letter "A" to identify points of equal height, grade or determine height difference on the earth.

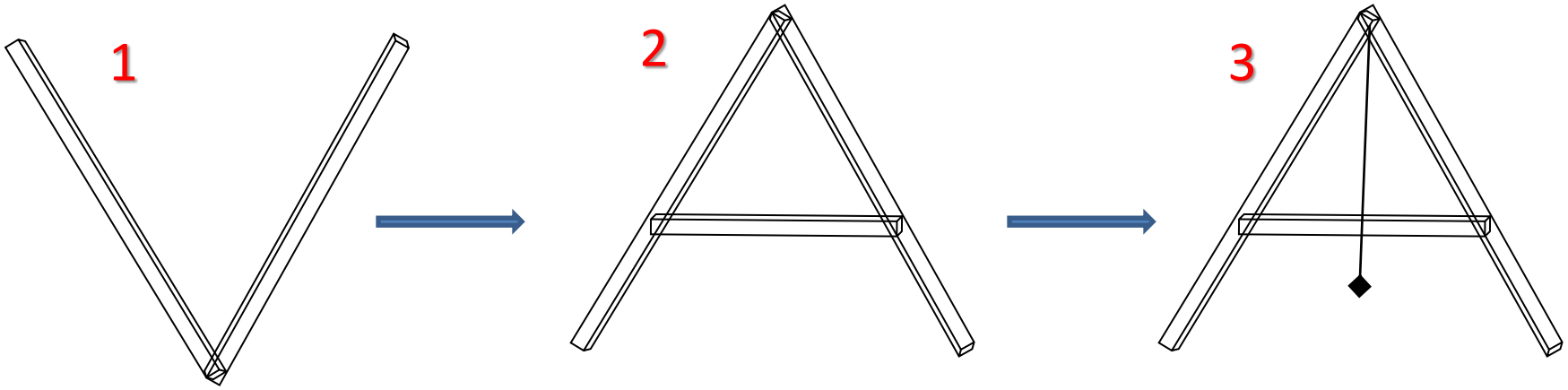
- **Uses**
  - Contour
  - Grade
  - Measure height difference

- **Materials required for an A-frame**
  - two identical wooden poles
  - cross bar
  - String
  - plummet or spirit level

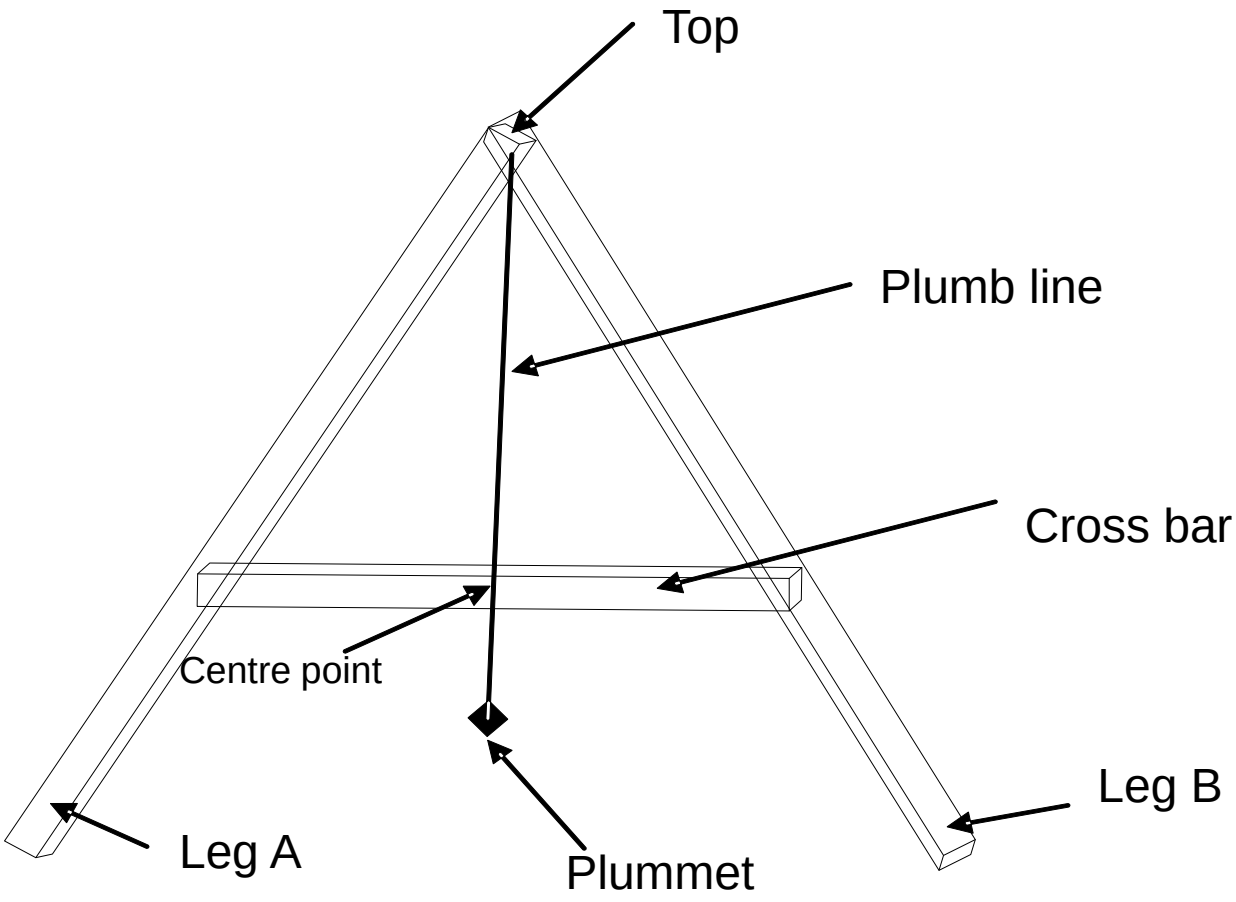
A-frame (2/11)

Construction of an A-frame

1. The poles are nailed together with nails to form a frame in the shape of a **V**.
2. A cross bar is used to hold them together in the form of **A**.
3. Tie one end of the string to the top of the “A” and the other end to the plummet (i.e. any weight (stone)) so as to hang freely just below the cross bar



A-frame (3/11)



A-frame (4/11)

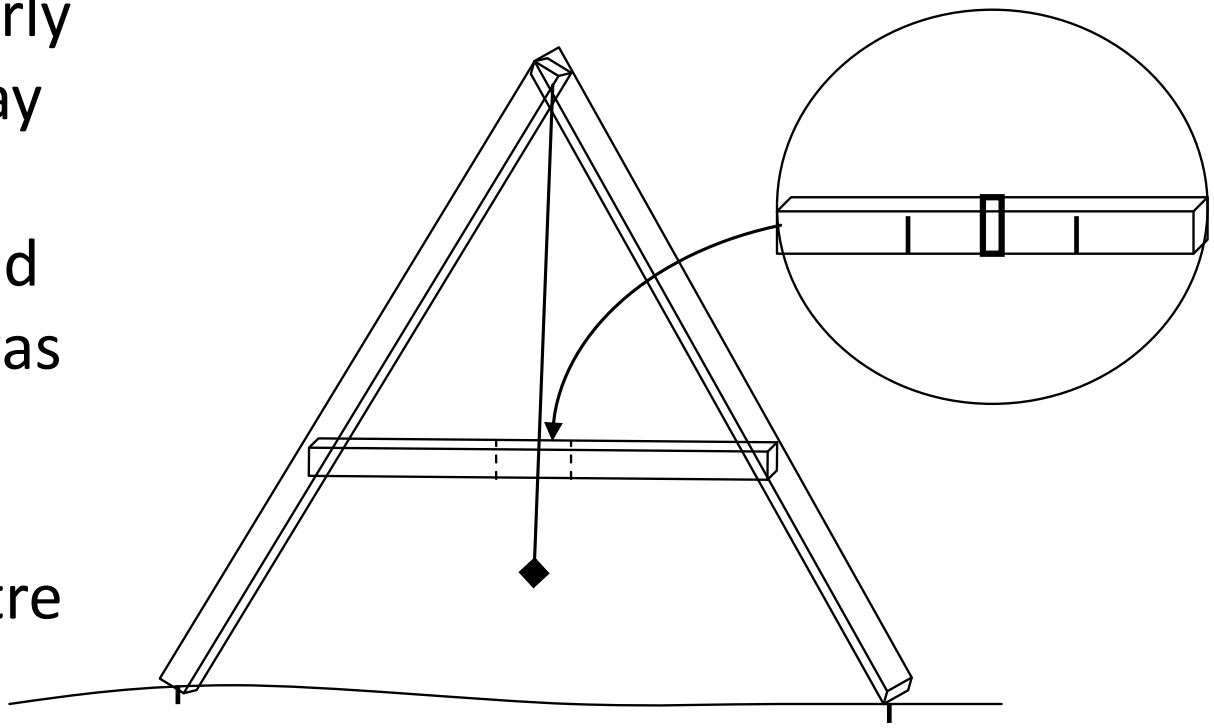
Calibrating the A-frame

1. Stand the A-frame upright on reasonably level ground. Mark on the ground where the two legs stand.
2. Hold the A-frame still, and use pencil to mark lightly on the crossbar where the string crosses it.
3. Turn the A-frame round, so that each leg stands exactly where the other had stood.
4. Make a second light mark on the crossbar where the string crosses it.

A-frame (5/11)

Calibrating the A-frame

5. The two marks on the crossbar should be fairly close together. Halfway between them gives where the string would cross if the A-frame was standing on perfectly level ground. Mark permanently this centre position with a heavy pencil mark or notch the bar with a knife.



A-frame (6/11)

Marking contour with the A-frame

1. Choose a place on the slope to start. Stand the A-frame up and mark where the first leg stands with a peg or stone.
2. Keeping the A-frame upright, and without moving the first leg, swing the second leg up or down the slope until the string crosses the crossbar exactly at the heavy pencil mark. Mark where the second leg stands on the ground with another peg or stone.
3. Keeping the second leg in the same place, lift the first leg up and pivot it around. Move it up and down the slope until you find the place where the string crosses the crossbar at the heavy pencil mark. Mark where the first leg is now standing with another peg or stone.
4. Continue in this manner to the end of the field.

A-frame (7/11)

Marking contour with the A-frame

5. The line of pegs or stones will mark a contour line i.e. they will all be at the same height on the slope. The pegs are usually not in a smooth curve. If necessary, make a smooth curve by moving them a little up or down.
6. To mark another contour line, move up or down the slope a certain distance—usually about 20 m (20 paces) on a gentle slope, or a drop of 1.5 m on steeper slopes. Repeat the process over the entire field of interest.
7. You can now dig ditches, construct terraces or plant trees along the contour lines using the pegs or stones as a guide.

A-frame (8/11)

To "grade" a field or a drainage canal the A-frame must be re-calibrated as follows:

1. Decide what percentage slope you want for the field or is required for the drainage canal.
2. Calculate the vertical/height difference. Example: 4% slope using an A-frame with legs 2 m apart, the VD is 8 cm.
3. Cut a small block of wood 8 cm height. Place the block on a level ground against a wall so that it does not move.

A-frame (9/11)

To "grade" a field or a drainage canal :

4. Place a straight 2-metre-long board or pole on top of the block, with the other end resting on the ground.
5. Place at the end of the board a heavy stone so that the board does not move or slide down. Do not stand on the board when marking the A-frame.
6. Place the A-frame on the board, and mark where the string passes on the crossbar of the A-frame. The A-frame is now calibrated; when the string aligns with the mark on the crossbar, which indicates a 4% slope.

A-frame (10/11)

Table: Block height and corresponding percentage slope for 2 m wide A-frame

Block height (cm)	Slope (%)
2	1
4	2
6	3
8	4
10	5
12	6

A-frame (11/11)

Using the recalibrated A-frame to mark out the slope of a field or drainage ditch

- Follow the same procedure as described for marking a contour.
- Move one of the A-frame legs up and down the field until the string passes through the 4% mark on the crossbar.

MAP SCALE AND SCALING

Learning Objectives

Students to be able to:

1. Read, determine and understand map scales
2. Transfer map distances to ground distance and vice versa

Definition

The *scale* of a *map* is the ratio of a distance on the *map* to the corresponding distance on the ground

or

Scale is the reduced distance on a map representing the actual distance on the ground or field.

Expressing Map Scale

Map scales may be expressed in three forms:

1. Words – eg. 'one millimetre to ten thousand millimetres or ten metres' ,
2. Ratio – eg. 1:10,000
3. Fraction – eg. 1/10,000

Map Scale

Table : Scale and map/ground measurement relationship

	Scale: 1/50,000	Scale: 1/100,000
1 mm on map ≡ on ground	50 m	100 m
1 km on ground ≡ map	20 mm	10 mm
1 ha on ground ≡ on map	2 mm × 2 mm	1 mm × 1 mm

Map Scale Sample Calculation

$$\frac{Distance\ on\ the\ map}{Distance\ on\ the\ ground} = \frac{2\ cm}{1\ km} =$$

$$\frac{2\ cm}{100{,}000\ cm} = \frac{20\ mm}{1{,}000{,}000\ mm} =$$

$$\frac{1}{50{,}000} = {}^{1}/_{50{,}000} = 1:50{,}000$$

Sample Questions

1. A map has been drawn to the scales given below, what will be the equivalent map distance of 10mm on the ground in meters
   I. 1:5,000
   II. 1:100,000
   III. 1:150,000
2. The distance of 2 km is to be represented on a map
   I. If the scale is 1:50,000 what will be the distance on a map
   II. If the scale is 1:250,000 what will be the distance on a map

Interpreting map Scale

- Map scale is often confused or interpreted incorrectly, because the smaller the map scale, the larger the reference number and vice versa.
- For example, a 1:100,000 scale map is considered a larger scale than a 1:250,000 scale map.

Map Scale interpretation

Scale level	Scale	Ground dist. of 1mm on map	Ground dist. of 1cm on map	Example of use
Largest	1:10000	10m	100m	Farm level planning
	1:25000 (Local scale)	25m	250m	Environmental impact assessment, agriculture,
	1:50000 (Local scale)	50m	500m	Transect walk, vegetation surveys, driving,
	1:100000 (Regional scale)	100m	1km	Planning, GPS navigation
	1:250000 (Reg. scale)	250m	2.5km	Route planning, mineral exploration
	1:1 million	1km	10km	General reference, wall map
Smallest	1:10 million	10km	100km	General reference, wall map

Determination of Scale for mapping

- $$y - axis = \frac{Max\ ground\ distance\ in\ y\ direction}{Max\ paper\ distance\ in\ y\ direction}$$

- $$x - axis = \frac{Max\ ground\ distance\ in\ x\ direction}{Max\ paper\ distance\ in\ x\ direction}$$

- Choose the smaller scale i.e. the one with larger reference value and round off to a higher value and use as scale

Assignment 1

1. List and describe how you will construct the following
   - N-frame (Odd number groups; 1, 3, 5…/surname starting with A)
   - Liquid level (Even number groups; 2, 4, 6… surname starting with B-Z)
2. Describe how you will use the constructed instrument to mark contour and grade a field
   - N-frame (Odd number groups; 1, 3, 5…/surname starting with A)
   - Liquid level (Even number groups; 2, 4, 6… surname starting with B-Z)

Quiz

1. List the material required to construct

   N-frame or

   Liquid level

2. Describe how you will use the constructed N-frame or Liquid level to mark contour a field

Assignment 2

1. A map has been drawn to the scales given below, what will be the equivalent map distance of 3 mm on the ground in meters
   a. 1:10,000
   b. 1:25,000
   c. 1:150,000
2. The distances of 5 km is to be represented on a map
   a. If the scale is 1:25,000 what will be the distance on a map
   b. If the scale is 1:100,000 what will be the distance on a map

Quiz

1. A map has been drawn to the scales given below, what will be the equivalent map distance of 10mm on the ground in meters
   I. 1:10,000
   II. 1:25,000
   III. 1:200,000
2. The distance of 4 km is to be represented on a map
   I. If the scale is 1:40,000 what will be the distance on a map
   II. If the scale is 1:200,000 what will be the distance on a map

Quiz

1. The ground distances given below has been drawn on a map with an equivalent map distance of 1 mm. What is the scale of the map?
   a. 25 m
   b. 50 m
2. A map has been drawn with an equivalent map distance of 50 mm. What is the ground distance in km?
   a. If the scale is 1:100,000
   b. If the scale is 1:150,000
3. List the materials required to construct the following:
   - N-frame (Odd number groups; 1, 3, 5…)
   - Liquid level (Even number groups; 2, 4, 6…)